NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

RACE Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 72**

REVISION ASSINGMENT # 01 (VECTOR-3D-01) TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If ABCD is regular tetrahedron with length of any edge 'ℓ', then the minimum distance of any vertex from the opposite face is -

(A) $\sqrt{\frac{2}{3}}\ell$ (B) $\sqrt{\frac{3}{2}}\ell$ (C) $\frac{2}{3}\ell^2$ (D) $\frac{1}{\sqrt{3}}\ell$

2. Two lines whose equations are $L_1 : \frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{\lambda}$ and $L_2 : \frac{x-2}{3} = \frac{y-3}{2} = \frac{z-2}{3}$ lie in the same plane. If L_1 intersects a plane $x + y + z = 15$ at P, then distance of P from (3, 4, 3) is -

(A) 6 (B) 4 (C) 2 (D) 3

3. Vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar vectors such that $|\vec{a}| = 2, |\vec{b}| = 3$ & $\vec{a} \wedge \vec{b} = 30°$. A unit vector $\vec{d}$ is perpendicular to all of them. If $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \frac{\hat{i}}{6} - \frac{\hat{j}}{3} + \frac{\hat{k}}{3}$, then $|\vec{c}.\hat{i}| + |\vec{c}.\hat{j}| + |\vec{c}.\hat{k}|$ is equal to -

(A) $\frac{5}{3}$ (B) $\frac{5}{9}$ (C) $\frac{5}{12}$ (D) $\frac{5}{18}$

4. If two diagonals of two rhombus are represented by vectors $\vec{d}_1 = \vec{a} + 2\vec{b}$, $\vec{d}_2 = 2\vec{a} - \vec{b}$ and $\vec{d}_3 = 2\vec{a} + \vec{b}$, $\vec{d}_4 = \vec{a} - 2\vec{b}$ respectively, then $[\vec{a}\ \vec{b}\ \vec{a} \times 2\vec{b}]$ equals :-

(A) $\frac{|\vec{a}|^2 |\vec{b}|^2}{4}$ (B) $\frac{|\vec{a}|^2 |\vec{b}|^2}{2}$ (C) $2|\vec{a}|^2 |\vec{b}|^2$ (D) $4|\vec{a}|^2 |\vec{b}|^2$

5. Let $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$ and $\vec{c} = \alpha\vec{a} + \beta\vec{b}$. If the vectors, $\hat{i} - 2\hat{j} + \hat{k}$, $3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c}$ are coplanar then $\frac{\alpha}{\beta}$ is -

(A) 1 (B) 2 (C) 3 (D) – 3

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Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

6. A plane meets the coordinate axes in A, B, C such that the centroid of the triangle ABC is the point $(1, r, r^2)$. The plane passes through the point (4, –8, 15) if r is equal to -

(A) –3 (B) 3 (C) 5 (D) –5

7. A non-zero vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j}, \hat{i}-\hat{k}$,. The possible angle between $\vec{a}$ and $\hat{i}-2\hat{j}+2\hat{k}$ is -

(A) $\pi/3$ (B) $\pi/4$ (C) $\pi/6$ (D) $3\pi/4$

8. If unit vectors $\hat{i}$ & $\hat{j}$ are at right angles to each other and $\vec{p}=3\hat{i}+4\hat{j}$, $\vec{q}=5\hat{i}$, $4\vec{r}=\vec{p}+\vec{q}$ and $2\vec{s}=\vec{p}-\vec{q}$, then -

(A) $|\vec{r}+k\vec{s}|=|\vec{r}-k\vec{s}|$ for all real k (B) $\vec{r}$ is perpendicular to $\vec{s}$

(C) $\vec{r}+\vec{s}$ is perpendicular to $\vec{r}-\vec{s}$ (D) $|\vec{r}|=|\vec{s}|=|\vec{p}|=|\vec{q}|$

9. If a vector $\vec{r}$ of magnitude $3\sqrt{6}$ is directed along the bisector of the angle between the vectors $\vec{a}=7\hat{i}-4\hat{j}-4\hat{k}$ & $\vec{b}=-2\hat{i}-\hat{j}+2\hat{k}$, then $\vec{r}$ is equal to -

(A) $\hat{i}-7\hat{j}+2\hat{k}$ (B) $\hat{i}+7\hat{j}-2\hat{k}$ (C) $-\hat{i}+7\hat{j}-2\hat{k}$ (D) $\hat{i}-7\hat{j}-2\hat{k}$

10. A line passes through two points A and B whose position vectors are $\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-3\hat{j}+\hat{k}$ respectively. The position vector of a point on it at a unit distance from the point A is -

(A) $\frac{1}{5}(5\hat{i}+\hat{j}-7\hat{k})$ (B) $\frac{1}{5}(4\hat{i}+9\hat{j}-13\hat{k})$ (C) $\frac{1}{5}(6\hat{i}+\hat{j}-7\hat{k})$ (D) $\frac{1}{5}(5\hat{i}+9\hat{j}-13\hat{k})$

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Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 11 to 13

A plane containing the lines $L_1 : 2x - y + 3z - 5 = 0 = x - 2y + 2z + 1$ and $L_2 : \frac{x-1}{\lambda-1} = \frac{y-2}{\lambda-2} = \frac{z-3}{\lambda-3}$.

If P be a variable point on the plane which is at a distance 4 from (0, 0, 0).

On the basis of above information answer the following :

11. Equation of the plane is -

(A) $x + y + z = 6$ (B) $2x - y + 3z - 9 = 0$ (C) $x - 2y + 2z - 3 = 0$ (D) can't be determined

12. The possible value of λ is -

(A) 0 (B) 1 (C) 2 (D) 3

13. Area of the region bounded by the locus of P is -

(A) π (B) 2π (C) 4π (D) 9π

SECTION–II

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors such that $|\vec{a}| = \sqrt{6}$, $|\vec{b}| = \sqrt{2}$, $|\vec{c}| = \sqrt{3}$ and a vector $\vec{v} = (\vec{a} \times (\vec{a} \times \vec{b})) \times \vec{c} + (\vec{b} \times (\vec{b} \times \vec{c})) \times \vec{a} + (\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b}$ where magnitude of $\vec{v}$ is M then $\frac{M^2}{100}$ is

2. Consider a line passing through the point P(3,4,5) & having direction ratios (2,–2,1). If image of point Q(1,–1,8) w.r.t. given line is Q', then length of median of $\Delta PQQ'$ through vertex P is -

3. If lines $bx = 1 - cy$, $bz = 1$ & $bx = 2 + ay$, $bz = 3 + cy$ are perpendicular, where a,b,c are sides of ΔABC, then minimum value of cos B is -

4. Let $\hat{a} + 2\hat{b} + \hat{c} = \hat{a} \times \hat{c}$ then $|\hat{a} \times \hat{b} + \hat{b} \times \hat{c} + \hat{a} + \hat{b} + \hat{c}|$ is equal to

5. Let $|\vec{a}| = 4$, $|\vec{b}| = 1$ and $\vec{b} \times \vec{\alpha} + \vec{a} = \vec{\alpha}$. If projection of $\vec{\alpha}$ along $\vec{b}$ is 3 then projection of $\vec{\alpha}$ along $\vec{a}$ is

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SECTION–IV

Matrix Match Type (One or More than one option correct)

For each entry in **Column-I** , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I		**Column-II**
(A)	Vectors $\vec{a},\vec{b},\vec{c}$ are of same magnitude when taken pairwise they makes equal angle. If $\vec{a}=\hat{i}+\hat{j}$, $\vec{b}=\hat{i}+\hat{k}$, then $\vec{c}$ is	(P)	(6, 16, 20)
		(Q)	(0, 1, 1)
(B)	If vector $\vec{r}$ satisfies $\vec{r}\times(2\hat{i}+\hat{j}+\hat{k})=\hat{j}-\hat{k}$, then $\vec{r}$ can be	(R)	(3, 1, –1)
(C)	If plane passes through (3, 4, 5) and meets coordinates axes in A, B, C, then point of intersection of plane through A, B, C parallel to coordinate planes can be	(S)	$\left(\frac{4}{3},-\frac{1}{3},-\frac{1}{3}\right)$
		(T)	(3, 2, 2)
(D)	$\vec{a}=\hat{i}+\hat{j}$, $\vec{b}=2\hat{i}-\hat{k}$, then point of intersection of line $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$ and $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ is		

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